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Power conversion apparatus

Abstract

A power conversion apparatus includes a power converter configured to convert DC power supplied from a battery into AC power by switching control, and is configured to supply the AC power from the power converter to a rotary electric machine having windings. Further, the power conversion apparatus includes a control unit that performs temperature-increase control of the battery by performing switching control of upper arm switches and lower arm switches so that a current flows to the battery via the power converter and the windings. In the temperature-increase control while the vehicle is stopped, the control unit controls the current so that a current value of the rotary electric machine gradually increases within a current control range in which a torque of the rotary electric machine is less than an upper limit torque.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. bypass application of International Application No. PCT/JP2022/011099 filed on Mar. 11, 2022, which designated the U.S. and claims priority to Japanese Patent Application No. 2021-066577 filed on Apr. 9, 2021, the contents of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to power conversion apparatus.

Description of the Related Art

(3) There have been known power conversion apparatuses that perform temperature-increase control of a storage battery by exchanging power between the storage battery and a capacitor via an inverter. For example, temperature-increase control of a storage battery is performed by causing a current to flow between a first storage battery and a second storage battery that constitute a battery pack via an inverter, windings of a rotary electric machine, and a connection circuit. The power conversion apparatus is assumed to be mounted in a vehicle, and the rotary electric machine may be, for example, a traveling motor.

SUMMARY

(4) According to the present disclosure, a power conversion apparatus is provided with a power converter having series connections of upper arm switches and lower arm switches and configured to convert DC power supplied from a battery into AC power by switching control, the power conversion apparatus being configured to supply the AC power from the power converter to a rotary electric machine having windings, the rotary electric machine being connected to an axle of a vehicle, the power conversion apparatus comprising: a control unit that performs temperature-increase control of the battery by performing switching control of the upper arm switches and the lower arm switches so that a current flows to the battery via the power converter and the windings, wherein, in the temperature-increase control while the vehicle is stopped, the control unit acquires a current value of the rotary electric machine flowing through a neutral point of the windings and controls the current so that the current value gradually increases within a current control range in which a torque of the rotary electric machine is less than a threshold value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects, features and advantages of the present disclosure will become apparent from the following detailed description with reference to the accompanying drawings. In the accompanying drawings:

- (2) FIG. 1 is a block diagram of a power conversion apparatus;
- (3) FIG. 2 is a flowchart showing temperature-increase control process;
- (4) FIG. 3 is a diagram showing an equivalent circuit;
- (5) FIG. 4 is a functional block diagram of a controller;
- (6) FIG. 5 is a diagram showing a method of setting a command current;
- (7) FIG. 6 is a timing diagram showing a transition of a control mode of a switch, and the like;
- (8) FIG. 7 is a diagram showing a simulation result;
- (9) FIG. 8 is a flowchart showing a command current setting process;
- (10) FIG. 9 is a timing diagram showing a transition of a current flowing through a neutral point and a torque;
- (11) FIG. 10 is a flowchart showing a command current setting process in a second embodiment;
- (12) FIG. 11 is a timing diagram showing a transition of a current flowing through a neutral point and a torque in the second embodiment;
- (13) FIG. 12 is a flowchart showing a command current setting process in a third embodiment;
- (14) FIG. 13 is a diagram for explaining phase current imbalance control;
- (15) FIG. 14 is a diagram schematically showing windings of respective phases and magnetic poles;
- (16) FIG. 15 is a diagram schematically showing windings of respective phases and magnetic poles;
- (17) FIG. 16 is a timing diagram showing a transition of a phase current and a torque in the third embodiment;
- (18) FIG. 17 is a flowchart showing a command current setting process in a fourth embodiment;

- (19) FIG. **18** is a timing diagram showing a transition of a current flowing through a neutral point and a torque in the fourth embodiment;
- (20) FIG. **19** is a flowchart showing command current setting process in a fifth embodiment;
- (21) FIG. **20** is a timing diagram showing a transition of a current flowing through a neutral point and a torque in the fifth embodiment;
- (22) FIG. **21** is a timing diagram showing a transition of a current flowing through a neutral point and a torque in a modified example; and
- (23) FIG. **22** is a timing diagram showing a transition of a current flowing through a neutral point and a torque in another modified example.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(24) There have been known power conversion apparatuses that perform temperature-increase control of a storage battery by exchanging power between the storage battery and a capacitor via an inverter. According to a patent literature, JP 2020-120566 A, temperature-increase control of a storage battery is performed by causing a current to flow between a first storage battery and a second storage battery that constitute a battery pack via an inverter, windings of a rotary electric machine, and a connection circuit. The power conversion apparatus is assumed to be mounted in a vehicle, and the rotary electric machine may be, for example, a traveling motor.

(25) During temperature-increase control while a vehicle is stopped, a large current supplied to windings of a rotary electric machine generates an unintended large motor torque, and the torque may exceed a parking brake holding torque, causing the vehicle to move.

First Embodiment

(26) With reference to the drawings, a first embodiment of a power conversion apparatus according to the present disclosure will be described below. In the present embodiment, the power conversion apparatus is mounted in a vehicle. It should be noted that the same or equivalent components in the following embodiments are denoted by the same reference signs in the drawings, and the description thereof will be omitted.

(27) As shown in FIG. **1**, a power conversion apparatus **10** includes an inverter **30** as a power converter connected to a rotary electric machine **40**. The power conversion apparatus **10** has a function of exchanging electrical power between a battery pack **20** and the rotary electric machine via the inverter **30** in order to raise the temperature of the battery pack **20** as a battery.

(28) The rotary electric machine **40** is a three-phase synchronous machine, and includes U-, V- and W-phase windings **41U**, **41V** and **41W** that are star-connected as stator windings. The phase windings **41U**, **41V** and **41W** are shifted from each other by an electrical angle of 120°. The rotary electric machine **40** may be, for example, a permanent magnet synchronous machine. In the present embodiment, the rotary electric machine **40** is a main onboard machine, and serves as a source of traveling power of the vehicle. That is, the rotary electric machine **40** is coupled to the axle.

(29) The inverter **30** includes three phases of series connections of upper arm switches QUH, QVH and QWH and lower arm switches QUL, QVL and QWL. In the present embodiment, voltage-controlled semiconductor switching elements are used as the switches QUH, QVH, QWH, QUL, QVL and QWL, and specifically, IGBTs, MOSFETs, and the like are used. Diodes DUH, DVH, DWH, DUL, DVL and DWL as freewheel diodes are connected to the switches QUH, QVH, QWH, QUL, QVL and QWL, respectively, in an antiparallel manner.

(30) A low potential terminal of the U-phase upper arm switch QUH and a high potential terminal of the U-phase lower arm switch QUL are connected to a first end of the U-phase winding **41U** of the rotary electric machine **40** via a U-phase conductive member **32U** such as a bus bar. A low potential terminal of the V-phase upper arm switch QVH and a high potential terminal of the V-phase lower arm switch QVL are connected to a first end of the V-phase winding **41V** of the rotary electric machine **40** via a V-phase conductive member **32V** such as a bus bar. A low potential terminal of the W-phase upper arm switch QWH and a high potential terminal of the W-phase lower arm switch QWL are connected to a first end of the W-phase winding **41W** of the rotary

electric machine **40** via a W-phase conductive member **32W** such as a bus bar. Second ends of the U-, V- and W-phase windings **41U**, **41V** and **41W** are connected to each other at a neutral point O. In the present embodiment, the phase windings **41U**, **41V** and **41W** have the same number of turns. Accordingly, the phase windings **41U**, **41V** and **41W** have the same inductance, for example.

(31) The high potential terminals of the upper arm switches QUH, QVH and QWH are connected to a positive electrode terminal of the battery pack **20** via a positive electrode bus Lp such as a bus bar. The low potential terminals of the lower arm switches QUL, QVL and QWL are connected to a negative electrode terminal of the battery pack **20** via a negative electrode bus Ln such as a bus bar.

(32) The power conversion apparatus **10** includes a capacitor (smoothing capacitor) **31** that connects the positive electrode bus Lp and the negative electrode bus Ln. The capacitor **31** may be incorporated in the inverter **30** or may be provided externally to the inverter **30**.

(33) The battery pack **20** is configured as a series connection of battery cells as unit cells, and may have a terminal voltage of several hundred volts, for example. In the present embodiment, the terminal voltages (for example, rated voltages) of the battery cells constituting the battery pack **20** are set to be the same. The battery cells may be, for example, secondary batteries such as lithium ion batteries.

(34) In the present embodiment, among the battery cells constituting the battery pack **20**, a series connection of a plurality of battery cells on the high potential side constitutes a first storage battery **21**, and a series connection of a plurality of battery cells on the low potential side constitutes a second storage battery **22**. That is, the battery pack **20** is divided into two blocks. In the present embodiment, the number of battery cells constituting the first storage battery **21** and the number of battery cells constituting the second storage battery **22** are the same. Accordingly, the terminal voltage (for example, rated voltage) of the first storage battery **21** and the terminal voltage (for example, rated voltage) of the second storage battery **22** are the same.

(35) In the battery pack **20**, the negative electrode terminal of the first storage battery **21** and the positive electrode terminal of the second storage battery **22** are connected to an intermediate terminal B.

(36) The power conversion apparatus **10** includes a monitoring unit **50** (corresponding to a voltage information detecting unit). The monitoring unit **50** monitors the terminal voltage, SOC, SOH, temperature, and the like of each battery cell constituting the battery pack **20**.

(37) The power conversion apparatus **10** includes a connection path **60** and a connection switch **61**. The connection path **60** electrically connects the intermediate terminal B of the battery pack **20** and the neutral point O. The connection switch **61** is provided on the connection path **60**. In the present embodiment, a relay is used as the connection switch **61**. When the connection switch **61** is turned on, the intermediate terminal B and the neutral point O are electrically connected to each other. On the other hand, when the connection switch **61** is turned off, the intermediate terminal B and the neutral point O are electrically isolated from each other.

(38) The power conversion apparatus **10** includes a current sensor **62** that detects a current flowing through the connection path **60** (that is, a current flowing through the neutral point). A value detected by the current sensor **62** is input to a controller **70** (corresponding to a control unit) of the power conversion apparatus **10**.

(39) The controller **70** is mainly configured with a microcomputer, and controls switching of each switch constituting the inverter **30** in order to perform feedback control of a controlled variable of the rotary electric machine **40** to a command value thereof. Thus, the power conversion apparatus **10** converts the DC power of the battery pack **20** into AC power and supplies it to the rotary electric machine **40**. The controlled variable may be, for example, torque.

(40) The controller **70** controls on/off of the connection switch **61**, and is capable of communicating with the monitoring unit **50**. Further, the controller **70** is capable of communicating with a high-order controller **80** provided external to the power conversion apparatus **10**. The high-order controller **80** performs overall control of the vehicle.

(41) The controller **70** implements various control functions by executing programs stored in its own storage device. The various functions may be implemented by an electronic circuit which is hardware, or may be implemented by both hardware and software.

(42) Next, description will be given of temperature-increase control for the battery pack **20** executed by the controller **70**. FIG. **2** is a flowchart showing a procedure of temperature-increase control process. This process is repeatedly executed by the controller **70** at a predetermined control cycle, for example.

(43) In step **S10**, it is determined whether there is a request to raise the temperature of the battery pack **20**. For example, when it is determined that there is an instruction to raise the temperature of the battery pack **20** from the high-order controller **80** or when it is determined that the temperature of the battery pack **20** detected by the monitoring unit **50** is less than a threshold temperature, it can be determined that there is a temperature-increase request. The temperature compared with the threshold temperature may be, for example, a lowest temperature among the detected temperatures of the battery cells or an average temperature of the battery cells calculated based on the detected temperatures of the battery cells. In the present embodiment, a situation affirmatively determined in step **S10** is assumed to be a situation in which the vehicle is stopped before the rotary electric machine **40** is driven.

(44) If it is determined that there is no temperature-increase request in step **S10**, the process proceeds to step **S11**, and it is determined whether there is a request to drive the rotary electric machine **40**. In the present embodiment, the drive request includes a request to make the vehicle travel by rotational driving of the rotary electric machine **40**.

(45) If it is determined that there is no drive request in step **S11**, the process proceeds to step **S12**, and a standby mode is set. By setting this mode, the switches QUH to QWL of the inverter **30** are controlled to be turned off. In step **S13**, the connection switch **61** is controlled to be turned off. Thus, the intermediate terminal B and the neutral point O are electrically isolated from each other.

(46) If it is determined that there is a drive request in step **S11**, the process proceeds to step **S14**, and a drive mode of the rotary electric machine **40** is set. In step **S16**, the connection switch **61** is controlled to be turned on. Thus, the intermediate terminal B and the neutral point O are electrically connected to each other via the connection path **60**. Then, in step **S16**, switching of the switches QUH to QWL of the inverter **30** is controlled to rotationally drive the rotary electric machine **40**. Thus, the drive wheels of the vehicle can rotate to make the vehicle travel. The switching control in step **S16** may be performed, for example, using PWM based on a comparison of magnitude between a command voltage applied to the phase windings **41U** to **41W** and a career signal (for example, a triangular wave signal), or a pulse pattern.

(47) If it is determined that there is a temperature-increase request in step **S10**, the process proceeds to step **S17**, and a temperature-increase control mode is set. In step **S18**, the connection switch **61** is controlled to be turned on. In step **S19**, temperature-increase PWM control is performed to raise the temperature of the battery pack **20**. This control will be described below.

(48) In FIG. **3**, chart (a) shows an equivalent circuit of the power conversion apparatus **10** used in temperature-increase PWM control. In chart (a) in FIG. **3**, the phase windings **41U** to **41W** are indicated as a winding **41**, the upper arm switches QUH, QVH and QWH are indicated as an upper arm switch QH, and the upper arm diodes DUH, DVH and DWH are indicated as an upper arm diode DH. Further, the lower arm switches QUL, QVL and QWL are indicated as a lower arm switch QL, and the lower arm diodes DUL, DVL and DWL are indicated as a lower arm diode DL.

(49) The equivalent circuit of chart (a) in FIG. **3** can be shown as an equivalent circuit of a chart (b) of FIG. **3**. The circuit of the chart (b) is a buck-boost chopper circuit capable of bidirectional power transmission between the first storage battery **21** and the second storage battery **22**. In the chart (b), VBH indicates the terminal voltage of the first storage battery **21**, IBH indicates the current flowing through the first storage battery **21**, VBL indicates the terminal voltage of the second storage battery **22**, and IBL indicates the current flowing through the second storage battery **22**. IBH and

IBL become negative when the charging current of the first and second storage batteries **21** and **22** flows, and positive when the discharging current of the first and second storage batteries **21** and **22** flows. Further, VR indicates the terminal voltage of the winding **41**, and IR indicates the current flowing through the neutral point O (corresponding to a current value of the rotary electric machine). IR becomes negative when the current flows through the neutral point O in the positive direction from the winding **41** to the intermediate terminal B, and positive when the current flows through the neutral point O in the opposite direction.

(50) With reference to FIG. **3**, chart (b), when the upper arm switch QH is turned on, the terminal voltage VR of the winding **41** becomes “VBH.” On the other hand, when the lower arm switch QL is turned on, the terminal voltage VR of the winding **41** becomes “-VBL.” That is, when the upper arm switch QH is turned on, an excitation current can flow through the winding **41** in the positive direction, and when the lower arm switch QL is turned on, an excitation current can flow through the winding **41** in the negative direction.

(51) FIG. **4** is a block diagram of temperature-increase PWM control. In the controller **70**, a current deviation calculation unit **71** calculates a current deviation by subtracting a current (hereinafter, referred to as a detected current IMr) detected by the current sensor **62** from a command current Im*. In the present embodiment, the command current Im* is set as a sinusoidal wave as shown in FIG. **5**. Specifically, the command current Im* is set so that a positive command current Im* and a negative command current Im* are point-symmetrical to a zero-crossing timing of the command current Im* in each cycle Tc of the command current Im*. Thus, a period from a zero-up crossing timing to a zero-down crossing timing of the command current Im* and a period from a zero-down crossing timing to a zero-up crossing timing of the command current Im* becomes equal to each other.

(52) Further, in each cycle Tc of the command current Im*, an area S1 of a first region and an area S2 of a second region becomes equal to each other. The first region S1 is a region surrounded by the time axis from the zero-up crossing timing to the zero-down crossing timing of the command current Im* and the positive command current Im* in each cycle Tc of the command current Im*. The second region is a region surrounded by the time axis from the zero-down crossing timing to the zero-up crossing timing of the command current Im* and the negative command current Im* in each cycle Tc. By setting “S1=S2”, it is possible to balance the charging and discharging currents of the first storage battery **21** and the second storage battery **22** in each cycle Tc, suppressing an increase in the difference between the terminal voltage of the first storage battery **21** and the terminal voltage of the second storage battery **22** due to the temperature-increase control.

(53) In addition, a frequency fc of the command current Im*, which is the reciprocal of a single cycle Tc of the command current Im*, may be preferably set to, for example, a frequency on the lower limit side of the human audible range. Specifically, the frequency fc is preferably set to 1 kHz or less, which is a frequency range in which the A-weighting correction value (dB) becomes or less, and more preferably set to a frequency between 30 Hz and 100 Hz (e.g., 50 Hz).

(54) A feedback control unit **72** calculates a duty ratio Duty as a manipulated variable for performing feedback control of the calculated current deviation to 0. The duty ratio Duty is a value that determines the ratio (Ton/Tsw) of the on-time Ton in a single switching cycle Tsw of each of the switches QUH to QWL. The feedback control used by the feedback control unit **72** may be, for example, proportional-integral control.

(55) A PWM generation unit **73** generates gate signals of the upper arm switches QUH, QVH and QWH based on the calculated duty ratio Duty. The gate signal is a signal for instructing on-control or off-control. In the present embodiment, the gate signals of the upper arm switches QUH, QVH and QWH are synchronized.

(56) An inverter **74** generates gate signals of the lower arm switches QUL, QVL and QWL by inverting the logic of the gate signals of the upper arm switches QUH, QVH and QWH generated by the PWM generation unit **73**. In the present embodiment, the gate signals of the lower arm

switches QUL, QVL and QWL are synchronized.

(57) FIG. 6 shows the transition of switching patterns and the like during temperature-increase PWM control. The chart (a) in FIG. 6 shows the transition of the gate signals of the upper arm switches QUH, QVH and QWH, and the chart (b) shows the transition of the gate signals of the lower arm switches QUL, QVL and QWL. The chart (c) shows the transition of a current I_R flowing through the neutral point O and the transition of the command current I_m^* . The chart (d) shows the transition of a current I_{BH} flowing through the first storage battery 21, and the chart (e) shows the transition of a current I_{BL} flowing through the second storage battery 22.

(58) As shown in FIGS. 6, charts (a) and 6(b), temperature-increase PWM control is performed in which the upper arm switches QUH, QVH and QWH and the lower arm switches QUL, QVL and QWL are controlled to be alternately turned on. This control is continued until there is no longer temperature-increase request in step S10 of FIG. 2. By this control, a pulsed current flows through the first storage battery 21 and the second storage battery 22 as shown in chart (d) and chart (e). During a period in which the command current I_m^* is positive, the first storage battery 21 is discharged and the second storage battery 22 is charged. On the other hand, during a period in which the command current I_m^* is negative, the second storage battery 22 is discharged and the first storage battery 21 is charged. Average values $I_{BH_{ave}}$ and $I_{BL_{ave}}$ of the pulsed currents are sinusoidal currents including a component of the same frequency as the frequency of the command current I_m^* .

(59) FIG. 7 shows the simulation results of the present embodiment. In FIG. 7, charts (a) to (c) correspond to the charts (c) to (e) in FIG. 6, respectively. Thus, a sinusoidal current flows through the first storage battery 21 and the second storage battery 22, increasing the temperature. The terminal voltage of the capacitor 31 does not change.

(60) By synchronizing the switching control, rotational driving of a rotor of the rotary electric machine 40 is suppressed. However, depending on the rotor position or the like, the rotor may be rotationally driven. In the present embodiment, it is assumed that the temperature-increase control is performed while the vehicle is stopped, so if a large current is suddenly supplied to the rotary electric machine 40, the rotor is rotationally driven, whereby the rotary electric machine 40 may output an unintended torque greater than or equal to a predetermined value, and the torque may exceed a parking brake holding torque, causing the vehicle to move.

(61) Therefore, in the present embodiment, the command current I_m^* is set as described below in order to control the magnitude of the current I_R flowing through the neutral point O. When starting the temperature-increase control, the controller 70 performs a command current setting process for determining the command current I_m^* shown in FIG. 8. The command current setting process is performed at a predetermined cycle during the temperature-increase control.

(62) When starting the command current setting process, the controller 70 determines whether command current $I_m^*(n) \geq \text{target value } I_{mref}$ (step S101). The command current $I_m^*(n)$ is the command current I_m^* in the current process, and “n” indicates the number of executions of the command current setting process. In the present embodiment, the command current $I_m^*(0)$, which is the initial value of the command current $I_m^*(n)$, is zero. The initial value may be arbitrarily changed as long as it is sufficiently smaller than the target value I_{mref} . Further, the target value I_{mref} is an amplitude command value of the current I_R desirable for raising the temperatures of the first storage battery 21 and the second storage battery 22, and is set within a current control range in which the torque of the rotary electric machine 40 is less than the upper limit torque. In the present embodiment, the target value I_{mref} corresponds to the upper limit value of the current control range. The upper limit torque is a torque that is set based on the parking brake holding torque capable of maintaining the vehicle in a stopped state, and is set to a value less than or equal to the parking brake holding torque.

(63) If the determination result in step S101 is affirmative, the controller 70 sets the target value I_{mref} as the command current $I_m^*(n+1)$ of the next process (step S102). After the process of step

S102, the controller 70 ends the command current setting process.

(64) On the other hand, if the determination result in step S101 is negative, the controller 70 sets a value obtained by adding an amplitude increase amount ΔI_m to the current command current I_m^* (n) as the command current $I_m^*(n+1)$ of the next process (step S103). Here, the amplitude increase amount ΔI_m refers to an amplitude increase amount per unit time (time change rate, increase rate), and is set at least to a value less than the upper limit torque. Preferably, the amplitude increase amount ΔI_m is set so that the command current I_m^* reaches the target value I_{mref} in a plurality of processes (for example, approximately 5 to 10 times) for preventing the rate of raising temperature from becoming too slow. After the process of step S103, the controller 70 ends the command current setting process.

(65) Thus, as shown in the chart (a) of FIG. 9, the (amplitude of) current IR flowing through the neutral point O gradually increases from a temperature-increase control start time T_O within the current control range determined by the upper limit torque. Accordingly, as shown in the chart (b) of FIG. 9, the torque of the rotary electric machine 40 also gradually increases but does not exceed the upper limit torque (indicated by the dotted line). That is, the torque can be prevented from exceeding the parking brake holding torque.

(66) According to the present embodiment detailed above, the following effects can be achieved.

(67) When starting temperature-increase control while the vehicle is stopped, the controller controls the current so that the current IR gradually increases within a current control range in which a torque of the rotary electric machine 40 is less than an upper limit torque. Thus, when the temperature-increase control is performed while the vehicle is stopped, a large current does not suddenly flow through the winding 41 of the rotary electric machine 40, preventing generation of an unintended large torque. Therefore, it is possible to prevent the torque from exceeding the parking brake holding torque, and prevent the vehicle from moving.

(68) The intermediate terminal B (corresponding to an intermediate point) and the neutral point are connected to each other via the connection path 60 without the switches QUIET to QWL of the inverter 30. With this configuration, the controller 70 performs switching control of the inverter so that a ripple current flows between the first storage battery 21 and the second storage battery 22 via the inverter 30, phase windings 41U, 41V and 41W and the connection path 60. This makes it possible to reduce the amount of fluctuation in the terminal voltage of the capacitor 31 without increasing the frequency $f_c (=1/T_c)$ of the reactive power (ripple current). Therefore, noise generated during temperature-increase control of the battery pack 20 can be reduced.

(69) Further, since the amount of fluctuation in the terminal voltage of the capacitor 31 can be reduced, the capacitance of the capacitor 31 can be reduced, and the capacitor 31 can be miniaturized.

(70) In the temperature-increase control, the controller 70 synchronizes the switching control of the upper arm switches QUH, QVH and QWH of all phases, and synchronizes the switching control of the lower arm switches QUL, QVL and QWL of all phases. Thus, each of the phase windings 41U, 41V and 41W can be regarded as an equivalent circuit in which the windings are connected in parallel. Therefore, the inductance of the winding during temperature-increase control can be reduced. Accordingly, the amount of change in the current flowing through the neutral point O in a single switching cycle T_{sw} can be increased, and temperature-increase control can be performed using a large current.

(71) Further, by synchronizing the switching control, rotational driving of the rotor of the rotary electric machine 40 can be suppressed.

(72) The controller 70 turns on the connection switch 61 when it is determined that there is a request to raise the temperature of the battery pack 20, and turns off the connection switch 61 when it is determined that there is no request to raise the temperature of the battery pack 20. Therefore, it is possible to prevent a current from flowing from the neutral point O to the intermediate terminal B when the vehicle is traveling.

Second Embodiment

(73) With reference to FIG. 10, command current setting process according to a second embodiment will be described. The second embodiment employs the basic configuration of the first embodiment, and the following description will be given focusing on configurations different from the first embodiment.

(74) When starting the command current setting process shown in FIG. 10, the controller 70 determines whether command current $I_m^*(n) \geq \text{target value } I_{mref}$, as in step S101 (step S201). If the determination result is affirmative, the controller 70 sets the target value I_{mref} as the command current $I_m^*(n+1)$ of the next process, as in step S102 (step S202). Then, the controller 70 ends the command current setting process.

(75) On the other hand, if the determination result in step S201 is negative, the controller 70 acquires a position $\theta(n)$ of the rotor of the rotary electric machine 40 detected by a position sensor (angle sensor, or the like) and determines whether it is the same as the previous position $\theta(n-1)$ of the rotor (step S203). That is, it is determined whether the rotor has remained stationary.

(76) If the determination result is affirmative, that is, if the rotor has not moved, the controller 70 sets a value obtained by adding an amplitude increase amount ΔI_m to the current command current $I_m^*(n)$ as the command current $I_m^*(n+1)$ of the next process, as in step S103 (step S204). Then, the controller 70 ends the command current setting process.

(77) On the other hand, if the determination result in step S203 is negative, that is, if the rotor has moved, the controller 70 sets a value obtained by subtracting an amplitude margin ΔI_{mm} from the current command current $I_m^*(n)$ as the command current $I_m^*(n+1)$ of the next process (step S205). The amplitude margin ΔI_{mm} is a predetermined positive value, and is set in advance to a value at which the rotor remains stopped. The amplitude margin ΔI_{mm} may be an arbitrary value, but is preferably a value approximately half the amplitude increase amount ΔI_m to a value equivalent to the amplitude increase amount ΔI_m . As a result, a value smaller than the current command current $I_m^*(n)$ by the amplitude margin ΔI_{mm} is set as the command current $I_m^*(n+1)$ of the next and subsequent processes.

(78) Thus, as shown in the chart (a) of FIG. 11, the (amplitude of) current I_R flowing through the neutral point O gradually increases from a temperature-increase control start time T_0 . Accordingly, as shown in the chart (b) of FIG. 11, the torque of the rotary electric machine 40 also gradually increases from the time T_0 .

(79) Then, at a time T_1 when the rotor has moved, the amplitude of the current I_R decreases by the amplitude margin ΔI_{mm} , and this state is maintained. Accordingly, the torque of the rotary electric machine 40 decreases at the time T_1 when the upper limit torque is reached, and this state is maintained.

(80) According to the present embodiment detailed above, the following effects can be achieved.

(81) When the rotor has moved, the controller 70 determines that the upper limit torque has been reached, and sets a value smaller than the current command current $I_m^*(n)$ by the amplitude margin ΔI_{mm} as the command current $I_m^*(n+1)$ of the next and subsequent processes so that the rotor does not move further. Thus, the vehicle can be prevented from moving, and the command current I_m^* can be increased as close to the upper limit torque as possible. This makes it possible to quickly raise the temperature.

(82) As described above, the amplitude increase amount ΔI_m is preferably set to a value as large as possible so that the rate of raising temperature does not become too slow. However, if it is too large, the torque may greatly exceed the upper limit torque, causing the vehicle to move. Therefore, when the rotor has moved, the controller 70 sets a value smaller than the current command current $I_m^*(n)$ by the amplitude margin ΔI_{mm} as the command current $I_m^*(n+1)$ of the next and subsequent processes. Thus, it is possible to set the amplitude increase amount ΔI_m to a value as large as possible to enable early temperature-increase while preventing the vehicle from moving forward.

Third Embodiment

(83) With reference to FIG. 12, command current setting process according to a third embodiment will be described. The third embodiment employs the basic configuration of the first embodiment, and the following description will be given focusing on configurations different from the first embodiment.

(84) In the third embodiment, the controller 70 is configured to be able to control the phase currents of the windings 41 of the respective phases of the rotary electric machine 40. That is, in the power conversion apparatus 10, the phase currents of the windings 41 of the respective phases of the rotary electric machine 40 (corresponding to a current value of the rotary electric machine in the third embodiment) are detected, and the detected values are compared with the command values of the phase currents to control the phase currents of the windings 41 of the respective phases of the rotary electric machine 40.

(85) When starting the command current setting process shown in FIG. 12, the controller 70 acquires command currents $I_u^*(n)$, $I_v^*(n)$ and $I_w^*(n)$ that command the amplitudes of the phase currents of the windings 41 of the respective phases (step S301). The command current $I_u^*(n)$ commands the amplitude of the phase current of the U-phase in the current process. The command current $I_v^*(n)$ commands the amplitude of the phase current of the V-phase in the current process. The command current $I_w^*(n)$ commands the amplitude of the phase current of the W-phase in the current process. Then, the controller 70 determines whether the command current $I_u^*(n)+I_v^*(n)+I_w^*(n) \geq \text{target value } I_{mref}$ (step S301). That is, it is determined whether the sum of the command currents $I_u^*(n)$, $I_v^*(n)$ and $I_w^*(n)$ is greater than or equal to the target value I_{mref} .

(86) If the determination result is affirmative, the controller 70 ends the command current setting process shown in FIG. 12. Then, the current command currents $I_u^*(n)$, $I_v^*(n)$ and $I_w^*(n)$ are used as the command currents $I_u^*(n+1)$, $I_v^*(n+1)$ and $I_w^*(n+1)$ of the next process.

(87) On the other hand, if the determination result in step S301 is negative, the controller 70 acquires a position $\theta(n)$ of the rotor of the rotary electric machine 40 detected by a position sensor and determines whether it is the same as the previous position $\theta(n-1)$ of the rotor (step S302). That is, it is determined whether the rotor has not moved.

(88) If the determination result in step S302 is affirmative, that is, if the rotor has not moved, the controller 70 sets values obtained by adding an amplitude increase amount $\Delta I_m/3$ to each of the command currents $I_u^*(n)$, $I_v^*(n)$ and $I_w^*(n)$ as the command currents $I_u^*(n+1)$, $I_v^*(n+1)$ and $I_w^*(n+1)$ of the next process (step S303). The amplitude increase amount $\Delta I_m/3$ is one-third of the amplitude increase amount ΔI_m described in the first embodiment.

(89) On the other hand, if the determination result in step S303 is negative, that is, if the rotor has moved, the controller 70 performs phase current imbalance control to determine the command currents $I_u^*(n+1)$, $I_v^*(n+1)$ and $I_w^*(n+1)$ of the next process (step S304).

(90) With reference to FIG. 13, the phase current imbalance control will be described. As shown in FIG. 13, the controller 70 determines the command currents $I_u^*(n+1)$, $I_v^*(n+1)$ and $I_w^*(n+1)$ of the next process based on the direction of the phase current of the windings 41 of the respective phases and a rotor position θ_e . The direction of the phase current refers to each of the cases where the phase current flows from the inverter 30 to the neutral point O as shown in FIG. 14 and where the phase current flows from the neutral point O to the inverter 30 as shown in FIG. 15. Further, as shown in FIG. 14, the rotor position θ_e refers to the position of a magnetic pole 90 (for example, N pole) of the rotor expressed in electrical angle with the position of the U-phase winding 41U taken as 0° (reference). This is merely an example, and the reference may be the position of any winding 41, and the magnetic pole may be S pole.

(91) For example, when the rotor position θ_e is 20° ($0^\circ \leq \theta_e < 30^\circ$) and the phase current flows from the inverter 30 to the neutral point O (inverter.fwdarw.neutral point), the current command current $I_u^*(n)$ is set as the next command current $I_u^*(n+1)$. Similarly, when the rotor position θ_e is 20° and the phase current flows from the inverter 30 to the neutral point O, a value obtained by adding the

amplitude increase amount $\Delta I_m/3$ to the current command current $I_v^*(n)$ is set as the next command current $I_v^*(n+1)$. Similarly, when the rotor position θ_e is 20° and the phase current flows from the inverter **30** to the neutral point O, a value obtained by subtracting the amplitude increase amount $\Delta I_m/3$ from the current command current $I_w^*(n)$ is set as the next command current $I_w^*(n+1)$.

(92) As a result, as shown in the diagram (b) of FIG. **14**, the magnitudes of the amplitudes of the phase currents become unbalanced between phases in the next and subsequent processes, and the torque of the rotary electric machine **40** decreases. The diagram (b) of FIG. **15** shows that the phase currents are unbalanced between phases when the rotor position θ_e is 20° ($0^\circ \leq \theta_e < 30^\circ$) and the phase current flows from the neutral point O to inverter **30** (neutral point.fwdarw.inverter).

(93) As shown in the chart (a) of FIG. **16**, by the command current setting process in the third embodiment, the (amplitude) of the phase current gradually increases from the control start time **T10**. Accordingly, as shown in the chart (b) of FIG. **16**, the torque of the rotary electric machine also gradually increases from the time **T10**.

(94) Then, at the time **T11** when the rotor has moved, the phase currents I_u , I_v and I_w are controlled to be unbalanced, and this state is maintained. Accordingly, the torque of the rotary electric machine **40** decreases at the time **T11** when the upper limit torque is reached, and this state is maintained. In FIG. **16**, the phase current I_u is indicated by the solid line, the phase current I_v is indicated by the dot-dash line, and the phase current I_w is indicated by the dot-dot-dash line.

(95) According to the present embodiment detailed above, the following effects can be achieved.

(96) When the rotor has moved, the controller **70** determines that the upper limit torque has been exceeded, and controls the magnitudes of the amplitudes of the phase currents of the windings **41** of the respective phases to be unbalanced so that the rotor does not move further. Specifically, the controller **70** acquires the rotor position θ_e and determines, based on the direction of the phase current of the windings **41** and the rotor position θ_e , the command currents $I_u^*(n+1)$, $I_v^*(n+1)$ and $I_w^*(n+1)$ of the next and subsequent processes whereby the phase currents of the windings **41** become unbalanced between phases. For example, based on the direction of the phase current and the rotor position θ_e , it is determined which of the three phases is increased in amplitude, which of the three phases is decreased in amplitude, and which of the three phases is maintained in amplitude. Then, the increase amount $\Delta I_m/3$ is added to the winding **41** whose amplitude is to be increased, and the amplitude increase amount $\Delta I_m/3$ is subtracted from the winding **41** whose amplitude is to be reduced. As a result, the magnitudes of the phase currents become unbalanced between phases, and the torque can be reduced. Since the same value is added or subtracted as the amplitude increase amount $\Delta I_m/3$, the total value of the command currents $I_u^*(n)$, $I_v^*(n)$ and $I_w^*(n)$ can be maintained. That is, the torque can be reduced without reducing the total value.

Therefore, the temperature can be raised more quickly.

Fourth Embodiment

(97) With reference to FIG. **17**, command current setting process according to a fourth embodiment will be described. The fourth embodiment employs the basic configuration of the first embodiment, and the following description will be given focusing on configurations different from the first embodiment.

(98) When starting the command current setting process shown in FIG. **17**, the controller **70** acquires vehicle attitude information, and determines the upper limit torque based on the vehicle attitude information (step **S400**). The vehicle attitude information is information for recognizing the gradient of the road on which a vehicle is stopped based on the inclination of the vehicle. For example, when the vehicle is stopped on a downward slope, the parking brake holding torque decreases due to the influence of the weight of the vehicle, compared with the case where the vehicle is stopped on a flat ground or an upward slope. That is, even when the torque of the rotary electric machine **40** is small, the vehicle is more likely to move forward. Therefore, in step **S400**, the upper limit torque is determined based on the vehicle attitude information.

(99) Specifically, the upper limit torque is reduced as the downward gradient increases. In the present embodiment, the same upper limit torque is set for the upward gradient as for the flat ground, but the upper limit torque may be increased for the upward gradient.

(100) Then, the controller **70** determines whether command current $I_m^*(n) \geq \text{target value } I_{mref}$, as in step **S101** (step **S401**). In the third embodiment, the current limit range may be changed depending on the upper limit torque, and the target value I_{mref} may be changed accordingly. That is, as the upper limit torque decreases, the target value I_{mref} may be reduced so that the torque of the rotary electric machine **40** does not exceed the upper limit torque.

(101) If the determination result in step **S401** is affirmative, the controller **70** sets the target value I_{mref} as the command current $I_m^*(n+1)$ of the next process, as in step **S102** (step **S402**).

(102) On the other hand, if the determination result in step **S401** is negative, the controller **70** sets a value obtained by adding an amplitude increase amount ΔI_m to the current command current $I_m^*(n)$ as the command current $I_m^*(n+1)$ of the next process (step **S403**). Here, the amplitude increase amount ΔI_m is set at least to a value less than the upper limit torque. Preferably, the amplitude increase amount ΔI_m is set so that the command current I_m^* reaches the target value I_{mref} in a plurality of processes (for example, approximately 5 to 10 times) for preventing the rate of raising temperature from becoming too slow. Therefore, the amplitude increase amount ΔI_m may be changed according to the target value I_{mref} .

(103) Thus, as shown in the chart (a) of FIG. **18**, the (amplitude of) current I_R flowing through the neutral point O gradually increases within the current control range. Accordingly, as shown in the chart (b) of FIG. **18**, the torque of the rotary electric machine **40** also gradually increases but does not exceed the upper limit torque. In addition, when the downward gradient is large (for example, 10%) compared with the gradient of 0% (upper limit torque is indicated by the dotted line), the upper limit torque is reduced (indicated by the dot-dash line).

(104) According to the present embodiment detailed above, the following effects can be achieved.

(105) The controller **70** acquires vehicle attitude information while the vehicle is stopped, and changes the upper limit torque based on the vehicle attitude information. Accordingly, the controller **70** sets the target value I_{mref} within the current limit range that does not exceed the upper limit torque. Thus, the controller **70** controls the current so that the current I_R gradually increases within the current control range in which the torque of the rotary electric machine **40** is less than the upper limit torque. Therefore, when the vehicle is stopped on a downward slope, the vehicle can be prevented from moving based on the temperature-increase control.

Fifth Embodiment

(106) With reference to FIG. **19**, command current setting process according to a fifth embodiment will be described. The fifth embodiment employs the basic configuration of the first embodiment, and the following description will be given focusing on configurations different from the first embodiment.

(107) When starting the command current setting process shown in FIG. **19**, the controller **70** determines an amplitude increase amount ΔI_{m1} and an amplitude decrease amount ΔI_{m2} (step **S500**). Specifically, the controller **70** may determine the amplitude increase amount ΔI_{m1} according to any of the state of the battery pack **20** (battery temperature, SOC), upper limit torque, current control range, target value I_{mref} , and the like, or a combination thereof. The amplitude increase amount ΔI_{m1} refers to an amplitude increase amount per unit time (time change rate, increase rate) as with the amplitude increase amount ΔI_m in the first embodiment. On the other hand, the amplitude decrease amount ΔI_{m2} refers to an amplitude decrease amount per unit time (time change rate, decrease rate). In the present embodiment, the amplitude decrease amount ΔI_{m2} is set to the same value as the amplitude increase amount ΔI_{m1} ($\Delta I_{m1} = \Delta I_{m2}$), but may be a different value. For example, the amplitude decrease amount ΔI_{m2} may be set to a value smaller than the amplitude increase amount ΔI_{m1} .

(108) The controller **70** determines whether command current $I_m^*(n) \geq \text{target value } I_{mref}$, as in step

S101 of the first embodiment (step **S501**).

(109) If the determination result in step **S501** is affirmative, the controller **70** determines whether command current $I_m^*(n) = \text{target value } I_{mref}$ (step **S501**). If the determination result in step **S502** is affirmative, the controller **70** sets the target value I_{mref} as the command current $I_m^*(n+1)$ of the next process, as in step **S102** (step **S503**). Then, the controller **70** ends the command current setting process.

(110) On the other hand, if the determination result in step **S502** is negative, the controller **70** sets a value obtained by subtracting the amplitude decrease amount Δi_{m2} determined in step **S500** from the current command current $I_m^*(n)$ as the command current $I_m^*(n+1)$ of the next process (step **S504**). Then, the controller **70** ends the command current setting process.

(111) On the other hand, if the determination result in step **S501** is negative, the controller **70** sets a value obtained by adding the amplitude increase amount Δi_{m1} determined in step **S500** to the current command current $I_m^*(n)$ as the command current $I_m^*(n+1)$ of the next process (step **S505**). Then, the controller **70** ends the command current setting process.

(112) Thus, as shown in the chart (a) of FIG. **20**, the increase rate (indicated by the dotted line) of the (amplitude of) current I_R flowing through the neutral point **O** can be arbitrarily changed.

Further, when the target value I_{mref} is exceeded, the decrease ratio of the (amplitude of) current I_R can be arbitrarily changed.

(113) According to the present embodiment detailed above, the following effects can be achieved.

(114) An appropriate amplitude increase amount Δi_{m1} can be set based on the state of the battery pack **20** (battery temperature, SOC), upper limit torque, current control range, target value I_{mref} , and the like. This makes it possible to quickly increase the current. Moreover, since an appropriate amplitude increase amount Δi_{m1} is set, it is possible to prevent the torque from exceeding the upper limit torque of the rotary electric machine **40**. Further, since the amplitude decrease amount Δi_{m2} can be set, the command current $I_m^*(n+1)$ of the next and subsequent processes can be quickly set below the target value I_{mref} if the target value I_{mref} has been exceeded.

(115) (Modified Examples of Above Embodiments)

(116) A part of the configuration in the above embodiments may be modified as below. The following description will be given of modified examples. Each of the above embodiments may be implemented in combination. In the above embodiments, the temperature-increase control may be performed by applying a direct current to the battery pack **20** as shown in FIG. **21**. Further, as shown in FIG. **22**, the temperature-increase control may be performed by applying a combination of a direct current and an alternating current to the battery pack **20**. Although the battery pack **20** is used in the above embodiments, a unit cell may be used. In the above embodiments, the temperature-increase PWM control may be performed by controlling on/off of two of three phases. In the above embodiments, the controller **70** may cause a current to flow between the capacitor **31** and the battery pack **20** via the inverter **30** and the winding **41**. In the phase current imbalance control of the fourth embodiment, the same amplitude increase amount $\Delta i_{m/3}$ is added or subtracted to unbalance the phase currents. However, the amplitude increase amount to be added and the amplitude increase amount to be subtracted may be different from each other as long as the phase currents become unbalanced between phases.

(117) Although the present disclosure has been described in accordance with the examples, the present disclosure should not be construed as limited to those examples or structures. The present disclosure encompasses various modifications and equivalent alterations. In addition, various combinations and forms, and other combinations and forms including only one element, one or more elements, or one or less elements are also within the scope and spirit of the present disclosure.

(118) (Conclusion)

(119) It is thus desired to provide a power conversion apparatus that does not cause a rotary electric machine to output a torque exceeding a predetermined value during temperature-increase control while a vehicle is stopped.

(120) In order to solve the above problem, a power conversion apparatus may be provided with a power converter having series connections of upper arm switches and lower arm switches and configured to convert DC power supplied from a battery into AC power by switching control, the power conversion apparatus being configured to supply the AC power from the power converter to a rotary electric machine having windings, the rotary electric machine being connected to an axle of a vehicle, the power conversion apparatus comprising: a control unit that performs temperature-increase control of the battery by performing switching control of the upper arm switches and the lower arm switches so that a current flows to the battery via the power converter and the windings, wherein, in the temperature-increase control while the vehicle is stopped, the control unit acquires a current value of the rotary electric machine flowing through a neutral point of the windings and controls the current so that the current value gradually increases within a current control range in which a torque of the rotary electric machine is less than a threshold value.

(121) Thus, in the temperature-increase control while the vehicle is stopped, the controller acquires a current value of the rotary electric machine and controls a current so that the current value gradually increases within a current control range in which a torque of the rotary electric machine is less than a threshold value. As a result, when the temperature-increase control is performed while the vehicle is stopped, a large current does not suddenly flow through the windings of the rotary electric machine, preventing generation of an unintended large torque. Therefore, it is possible to prevent the vehicle from moving.

Claims

1. A power conversion apparatus provided with a power converter having series connections of upper arm switches and lower arm switches and configured to convert DC power supplied from a battery into AC power by switching control, the power conversion apparatus being configured to supply the AC power from the power converter to a rotary electric machine having windings, the rotary electric machine being coupled to an axle of a vehicle, the power conversion apparatus comprising: a control unit that performs temperature-increase control of the battery by performing switching control of the upper arm switches and the lower arm switches so that a current flows to the battery via the power converter and the windings, wherein in the temperature-increase control while the vehicle is stopped, the control unit controls the current so that a current value of the rotary electric machine gradually increases within a current control range in which a torque of the rotary electric machine is less than an upper limit torque.
2. The power conversion apparatus according to claim 1, wherein the battery includes a first storage battery and a second storage battery connected in series to an intermediate point, the intermediate point and a neutral point of the windings are electrically connected to each other via a connection path, and the control unit performs switching control of the upper arm switches and the lower arm switches so that a current flows between the first storage battery and the second storage battery via the power converter, the windings and the connection path.
3. The power conversion apparatus according to claim 1, wherein when the current value is gradually increased and reaches the upper limit value of the current control range, the control unit controls the current value so that the current value becomes equal to the upper limit value or less than the upper limit value.
4. The power conversion apparatus according to claim 1, wherein the control unit acquires a position of a rotor of the rotary electric machine, and, when the position of the rotor has moved in the gradual increase of the current value, controls the current value so that the current value becomes less than or equal to the current value during operation.
5. The power conversion apparatus according to claim 1, wherein the control unit acquires vehicle attitude information regarding a attitude of the vehicle, and changes the upper limit torque according to the vehicle attitude information.

6. The power conversion apparatus according to claim 1, wherein the control unit causes an alternating current to flow through the windings.

7. The power conversion apparatus according to claim 1, wherein the control unit is configured to be able to change an increase rate or a decrease rate per unit time of the current value of the rotary electric machine.

8. The power conversion apparatus according to claim 1, wherein the rotary electric machine includes windings of a plurality of phases, and the control unit acquires a position of a rotor of the rotary electric machine, and, when the position of the rotor has moved, unbalances phase currents of the windings between phases, based on a direction of the phase currents of the windings and the position of the rotor.

9. The power conversion apparatus according to claim 8, wherein when unbalancing the phase currents of the windings between phases, the control unit increases an amplitude of at least one phase current of the phase currents of the plurality of phases while decreasing an amplitude of at least one phase current of the phase currents of the plurality of phases.
